

MOHAMED NOMEIR

GENERAL INFORMATION

Affiliation: University of Maryland, College Park
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RESEARCH INTERESTS

Quantum communication and quantum information theory; quantum-enhanced distributed information processing; classical and quantum coding; information-theoretic privacy and security; distributed computation and storage; communication and information theory.

EDUCATION AND QUALIFICATIONS

- **Ph.D. Candidate in Electrical and Computer Engineering, University of Maryland, College Park** Expected 2027
Advisor: Prof. Sennur Ulukus
- **M.Sc. in Electronics and Communications Engineering, American University in Cairo** June 2021
- **B.Sc. in Electrical Engineering, Alexandria University** June 2018

PUBLICATIONS

CONFERENCE PAPERS

1. **M. Nomeir**, A. Aytekin, L. Hu, and S. Ulukus, "Quantum Codes for Private Distributed Matrix Multiplication in the High Privacy Regime", MILCOM 2026.
2. L. Hu, **M. Nomeir**, A. Aytekin, and S. Ulukus, "Simultaneously Minimizing Storage and Bandwidth Under Exact Repair With Quantum Entanglement", ITW 2026.
3. **M. Nomeir**, S. Meel, and S. Ulukus, "Private Information Retrieval With Arbitrary Privacy Requirements for Graph-Based Storage", ITW 2026.
4. S. Meel, **M. Nomeir**, and S. Ulukus, "Local Private Information Retrieval: A New Privacy Perspective for Graph-Based Replicated Systems", ITW 2026.
5. **M. Nomeir**, A. Aytekin, L. Hu, and S. Ulukus, "Quantum Codes for Private Distributed Matrix Multiplication for the Low Privacy Regime", ITW 2026.
6. L. Hu, **M. Nomeir**, A. Aytekin, and S. Ulukus, "Breaking the Storage-Bandwidth Tradeoff in Distributed Storage with Quantum Entanglement", ISIT 2026.
7. **M. Nomeir** and S. Ulukus, "Storage-Rate Trade-off in A-XPIR", ISIT 2026.

8. A. Aytekin, **M. Nomeir**, L. Hu, and S. Ulukus, "Convergence Properties of Good Quantum Codes for Classical Communication", ISIT 2026.
9. **M. Nomeir**, A. Aytekin, L. Hu, and S. Ulukus, "Quantum GASP Codes for Private Distributed Matrix Multiplication", ICASSP 2026.
10. **M. Nomeir**, A. Aytekin, and S. Ulukus, "The Capacity of Semantic Private Information Retrieval with Colluding Servers", GLOBECOM 2025.
11. **M. Nomeir**, P. Dissanayake, S. Meel, S. Dutta, and S. Ulukus, "Private Counterfactual Retrieval", Allerton 2025.
12. S. Meel, P. Dissanayake, **M. Nomeir**, S. Dutta, and S. Ulukus, "Private Counterfactual Retrieval With Immutable Features", ISIT 2025.
13. **M. Nomeir**, A. Aytekin, and S. Ulukus, "Byzantine Server and Static Eavesdropper Coalition in a Quantum XTUSPIR System", ICC 2025.
14. L. Hu, **M. Nomeir**, A. Aytekin, Y. Shi, S. Ulukus, and S. Guha, "Entanglement-Assisted Coding for Arbitrary Linear Computations Over a Quantum MAC", ITW 2025.
15. **M. Nomeir**, A. Aytekin, and S. Ulukus, "The Asymptotic Capacity of Byzantine Symmetric Private Information Retrieval and Its Consequences", ISIT 2025.
16. **M. Nomeir**, A. Aytekin, and S. Ulukus, "Quantum X -Secure B -Byzantine T -Colluding Private Information Retrieval", ITW 2024.
17. A. Aytekin, **M. Nomeir**, and S. Ulukus, "Quantum Private Membership Aggregation", ISIT 2024.
18. **M. Nomeir**, S. Vithana, and S. Ulukus, "Asymmetric X -Secure T -Private Information Retrieval: More Databases Are Not Always Better", CISS 2024.
19. **M. Nomeir**, S. Vithana, and S. Ulukus, "Private Membership Aggregation", MILCOM 2023.
20. A. Aytekin, **M. Nomeir**, S. Vithana, and S. Ulukus, "Quantum Symmetric Private Information Retrieval with Secure Storage and Eavesdroppers", GLOBECOM 2023.
21. **M. Nomeir**, Y. Gadallah, and K. G. Seddik, "Uplink Scheduling for Mixed Grant-Based eMBB and Grant-Free URLLC Traffic in 5G Networks", WiMob 2021.

JOURNAL ARTICLES AND MANUSCRIPTS

1. **M. Nomeir**, S. Meel, and S. Ulukus, "Private Information Retrieval With Arbitrary Privacy Requirements: Introduction and Capacity Results", submitted.
2. S. Meel, **M. Nomeir**, and S. Ulukus, "Local Private Information Retrieval for Graph-Based Replicated Systems", submitted.
3. L. Hu, **M. Nomeir**, A. Aytekin, and S. Ulukus, "Distributed Storage Systems with Quantum Entanglement Assistance", submitted.
4. **M. Nomeir**, A. Aytekin, L. Hu, and S. Ulukus, "Quantum Private Distributed Matrix Multiplication: New Families of Quantum Codes", submitted.

5. **M. Nomeir**, A. Aytekin, L. Hu, and S. Ulukus, "Quantum Private Distributed Matrix Multiplication: Extending the Classical Codes and Limitations", submitted.
6. S. Meel, **M. Nomeir**, P. Dissanayake, S. Dutta, and S. Ulukus, "What If, But Privately: Private Counterfactual Retrieval", IEEE Transactions on Information Theory, 2026.
7. **M. Nomeir**, A. Aytekin, and S. Ulukus, "Byzantine-Eavesdropper Alliance: How to Achieve Symmetric Privacy in Quantum X -Secure B -Byzantine E -Eavesdropped U -Unresponsive T -Colluding PIR?", IEEE Transactions on Information Theory, 2026.
8. A. Aytekin, **M. Nomeir**, and S. Ulukus, "Quantum X -Secure B -Byzantine T -Colluding Private Information Retrieval", IEEE Transactions on Information Theory, 2026.
9. A. Aytekin, **M. Nomeir**, S. Vithana, and S. Ulukus, "Quantum X -Secure E -Eavesdropped T -Colluding Symmetric Private Information Retrieval", IEEE Transactions on Information Theory, 2025.
10. **M. Nomeir**, Y. Gadallah, and K. G. Seddik, "Machine Learning-Based Uplink Scheduling Approaches for Mixed Traffic in Cellular Systems", IEEE Access, 2023.

TEACHING ASSISTANT

AMERICAN UNIVERSITY IN CAIRO (AUC):

- ECNG2105: Circuit Analysis I
- ECNG3201: Signals and Systems
- ECNG4302: Fundamentals of Communications II
- ECNG4306: Computer Communication Networks
- ECNG4308: Telecommunications Systems

UNIVERSITY OF MARYLAND, COLLEGE PARK (UMD):

- ENEE 222: Elements of Discrete Signal Analysis (Spring 2022–Spring 2023)
- ENEE 662: Convex Optimization (Fall 2023)

RESEARCH ASSISTANT

- **Bridging the Communications Gap: An AI-Based UAV Ubiquitous 5G Communication System** (September 2021–January 2022)
- **Wireless-Controlled Fully Implantable Smart Micro Pumping System for Local Anti-Cancer Drug Delivery in Pediatric Patients** (July 2020–May 2021)
- **Machine Learning Algorithms in 4G Systems** (August 2018–February 2019)

MASTER'S THESIS

Title: "Machine Learning-Based Critical Resource Allocation in Mixed-Traffic Cellular Networks."

Advisors: Prof. Yasser Gadallah and Prof. Karim Seddik.

Developed mathematical and machine-learning frameworks for uplink scheduling of grant-based eMBB and grant-free URLLC traffic in 5G networks.

B.SC. GRADUATION PROJECT

Title: "Security Aspects of Wireless Communications."

Advisor: Prof. Said El-Khamy.

Studied jamming and anti-jamming techniques, A5/1 stream-cipher security, and physical-layer security using coset coding with LDPC codes.

FELLOWSHIPS AND AWARDS

- *Medhat Haroun Award for Scholarly Excellence*, Best Engineering Thesis, American University in Cairo, 2021.
- *GTA Fellowship and Dean's Fellowship*, University of Maryland.
- *University Fellowship and Laboratory Fellowship*, American University in Cairo.
- *Top 10 Posters Award*, poster entitled "GSM System Attacks using Open-BTS Applications", NSRC 2018, Cairo.

PROFESSIONAL SERVICE

Reviewer

- International Symposium on Information Theory (ISIT)
- Information Theory Workshop (ITW)
- Symposium on Discrete Algorithms (SODA)
- IEEE Transactions on Information Theory (TIT)
- IEEE Transactions on Communications (TCOM)
- IEEE Journal on Selected Areas in Information Theory (JSAT)
- IEEE Transactions on Information Forensics and Security (TIFS)
- IEEE Open Journal on Communications (OJCOMS)
- Arabian Journal for Science and Engineering (AJSE)
- Frontiers of Information Technology & Electronic Engineering